Technology Stack

Date	1 July 2026
Team ID	RW-2026-01
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

The Rising Waters system integrates a responsive web interface with a Flask-based backend and an **XGBoost** machine learning model to provide accurate flood predictions. Data preprocessing and analysis are performed using Python libraries, while GitHub manages source code and collaboration. This technology stack ensures a scalable, secure, and reliable flood prediction system capable of supporting disaster management and early warning services.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, JavaScript	Provides a responsive, user-friendly interface for displaying flood predictions, alerts, and dashboards across desktop and mobile devices.

2	Backend / Server-Side	Python, Flask	Flask offers a lightweight and efficient framework for integrating the machine learning model, handling requests, and processing prediction results.
3	Database / Data Storage	CSV Dataset, Joblib Model Files (floods.save, transform.save)	Stores historical flood data for model training and saves the trained machine learning model and scaler for fast prediction during deployment.
4	Cloud / Hosting / Deployment	Local Flask Server / Render / Railway / AWS (Optional)	Enables easy deployment, scalability, and remote accessibility of the flood prediction application.
5	Version Control & CI/CD	Git, GitHub	Supports version control, team collaboration, project tracking, and continuous integration for efficient software development.
6	Third-Party APIs / Other Tools	Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Jupyter Notebook	Used for data preprocessing, visualization, machine learning model development, evaluation, and performance analysis.